DevOps- Assignment-3

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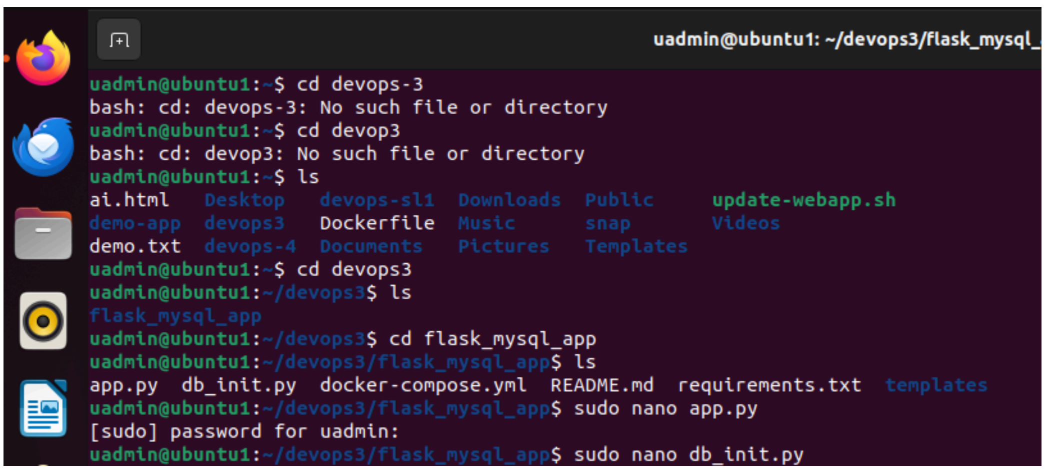
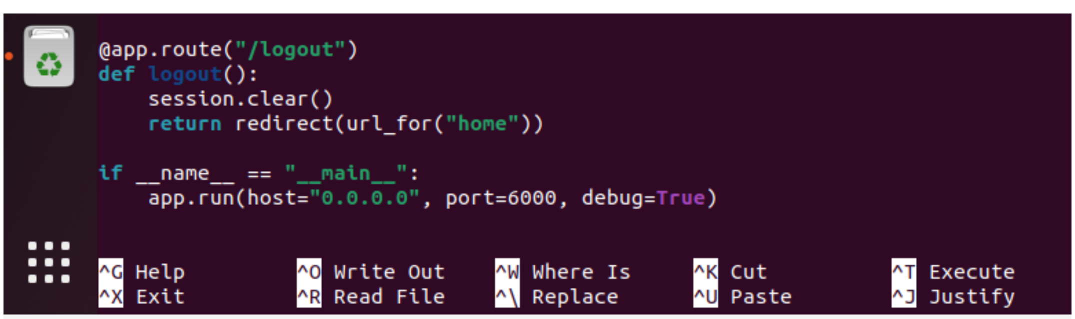
Configure the given flask application to run on port 6000.
Create a docker image of this application.
Create a repository on docker hub and push the above image to the docker repository.
Deploy the application on minikube.
Create a service for the deployment and check if it displays the application on the service URL.

Now modify the flask application.
Create a version2 docker image of this application.
Push the docker image to the above repository with a different tag.
On minikube set the new image for the deployment so that when you refresh the earlier page and make sure it is showing the new page.

1. Configuring the flask app





2. Creating a docker image of this application.

Dockerfile

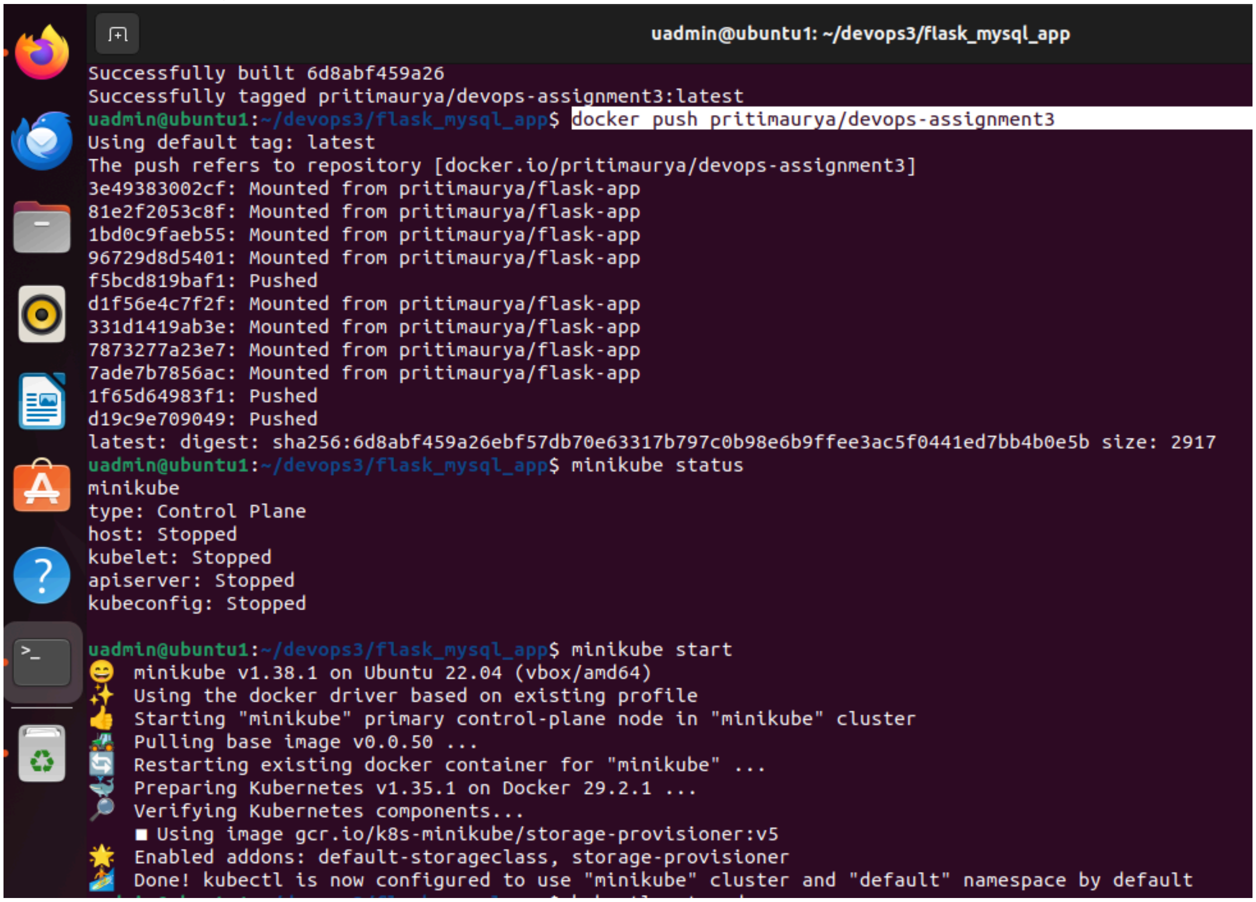
```
uadmin@ubuntu1: ~/devops3/flask_mysql_app
GNU nano 6.2                                        Dockerfile
FROM ubuntu
RUN apt update -y
RUN DEBIAN_FRONTEND=noninteractive apt install python3 -y
RUN apt install python3-pip python3-venv -y
RUN python3 -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip3 install flask
RUN pip3 install mysql-connector-python
RUN mkdir /app
COPY app.py /app/
COPY templates /app/templates
CMD ["python3", "/app/app.py"]
```

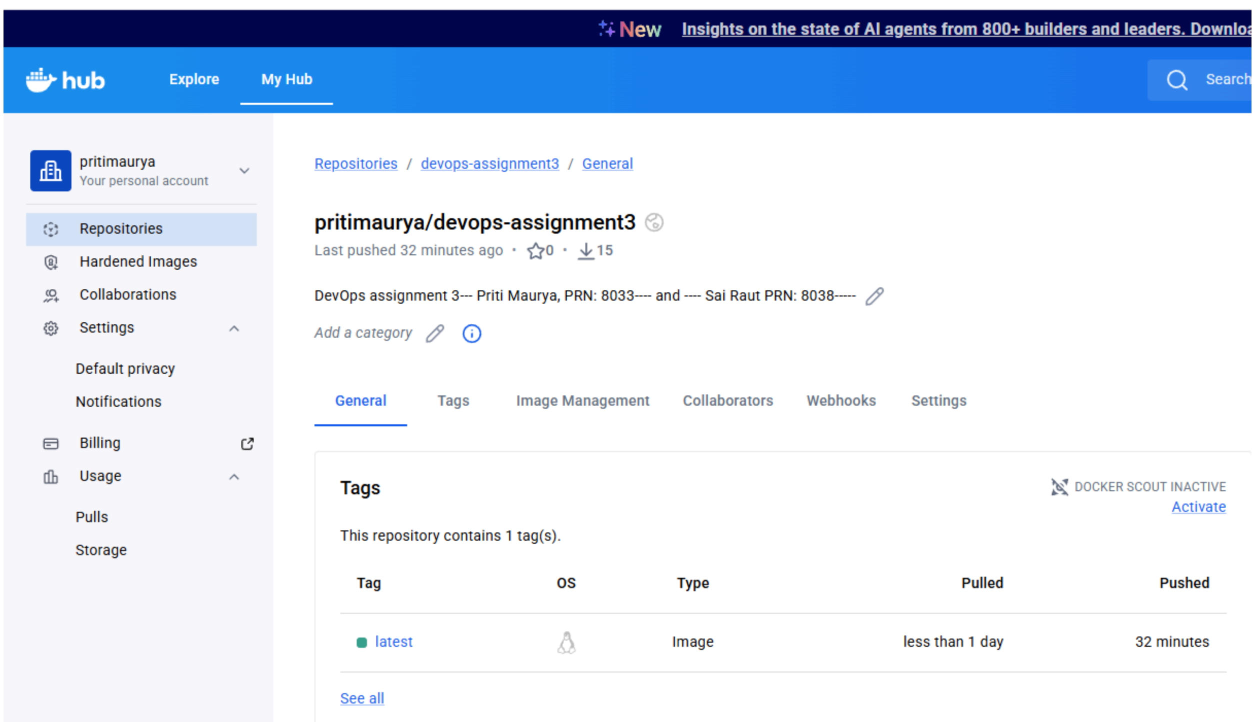
```
uadmin@ubuntu1: ~/devops3/flask_mysql_app
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker build -t pritimaurya/devops-assignment3 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  14.34kB
Step 1/12 : FROM ubuntu
 ---> 53958ec7b67c
Step 2/12 : RUN apt update -y
 ---> Using cache
 ---> 4332bf99b4c9
Step 3/12 : RUN DEBIAN_FRONTEND=noninteractive apt install python3 -y
 ---> Using cache
 ---> f0aa1e873e25
Step 4/12 : RUN apt install python3-pip python3-venv -y
 ---> Using cache
 ---> 8508944bb250
Step 5/12 : RUN python3 -m venv /opt/venv
 ---> Using cache
 ---> 156cdb56cf89
Step 6/12 : ENV PATH="/opt/venv/bin:$PATH"
 ---> Using cache
 ---> 412a2d3c72bc
Step 7/12 : RUN pip3 install flask
 ---> Using cache
 ---> b4436628a668
Step 8/12 : RUN pip3 install mysql-connector-python
 ---> Using cache
 ---> 09d435f420f9
Step 9/12 : RUN mkdir /app
 ---> Running in 12e4920f0cb9
 ---> Removed intermediate container 12e4920f0cb9
 ---> d0370b741e22
Step 10/12 : COPY app.py /app/
 ---> 1e4e9cdb2515
Step 11/12 : COPY templates /app/templates
 ---> f86580bbbd7c
Step 12/12 : CMD ["python3", "/app/app.py"]
 ---> Running in 580d299fa1e3
 ---> Removed intermediate container 580d299fa1e3
```

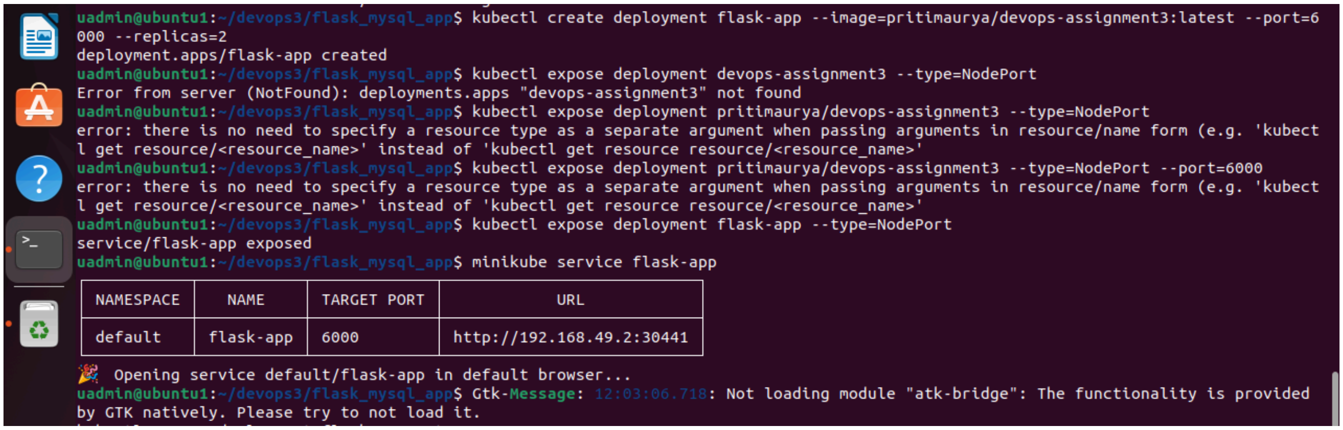
```
 ---> 6d8abf459a26
Successfully built 6d8abf459a26
Successfully tagged pritimaurya/devops-assignment3:latest
```

3. Create a repository on docker hub and push the above image to the docker repository.

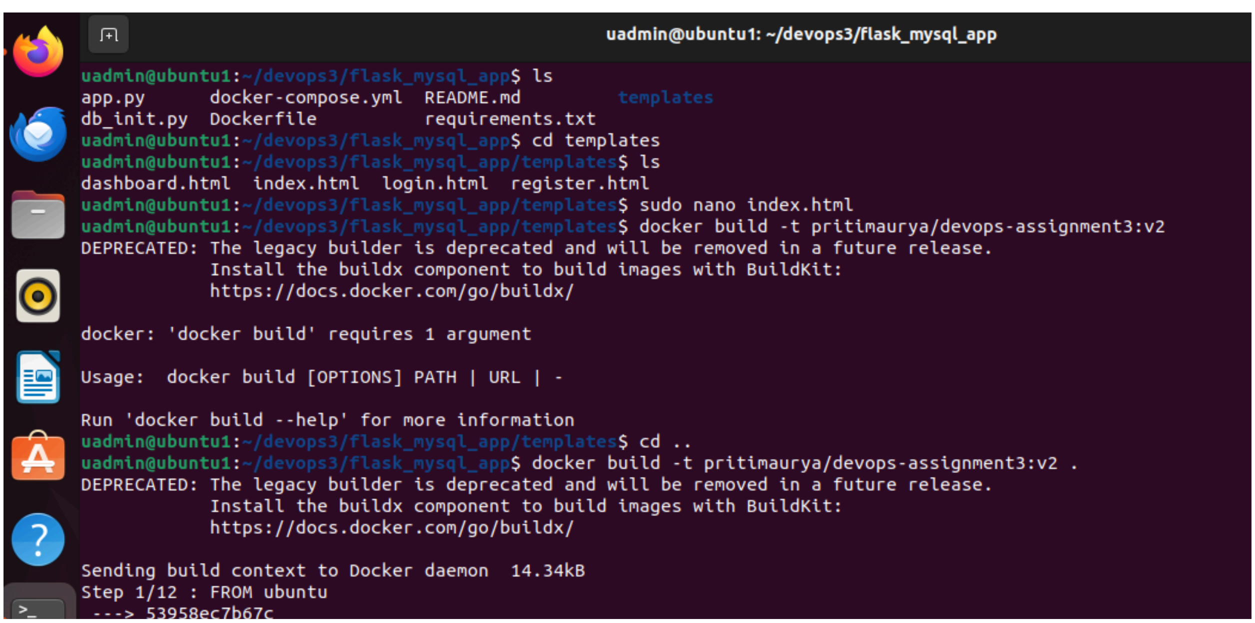




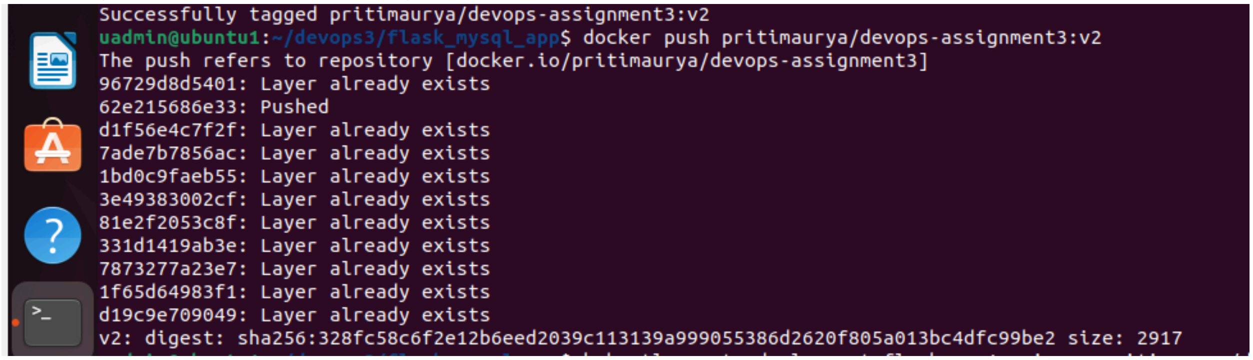
4. Deploy the application on minikube:
   - Creating service for the deployment



5. Now modifying the flask application,
   -Creating a version2 docker image of this application

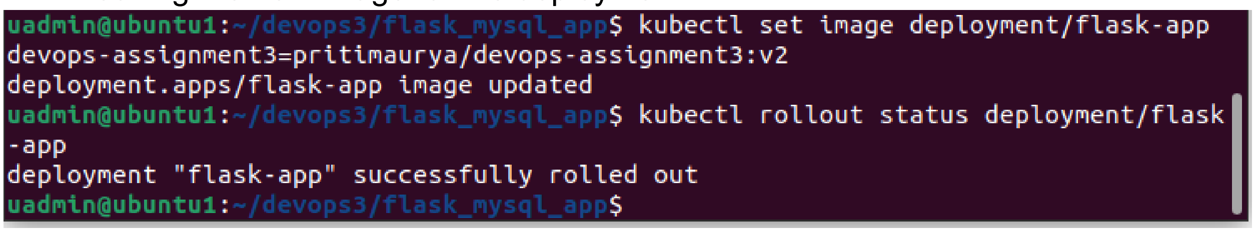


6. Pushing the new docker image



```
Successfully tagged pritimaurya/devops-assignment3:v2
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker push pritimaurya/devops-assignment3:v2
The push refers to repository [docker.io/pritimaurya/devops-assignment3]
96729d8d5401: Layer already exists
62e215686e33: Pushed
d1f56e4c7f2f: Layer already exists
7ade7b7856ac: Layer already exists
1bd0c9faeb55: Layer already exists
3e49383002cf: Layer already exists
81e2f2053c8f: Layer already exists
331d1419ab3e: Layer already exists
7873277a23e7: Layer already exists
1f65d64983f1: Layer already exists
d19c9e709049: Layer already exists
v2: digest: sha256:328fc58c6f2e12b6eed2039c113139a999055386d2620f805a013bc4dfc99be2 size: 2917
```

7. setting the new image for the deployment



```
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl set image deployment/flask-app
devops-assignment3=pritimaurya/devops-assignment3:v2
deployment.apps/flask-app image updated
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl rollout status deployment/flask
-app
deployment "flask-app" successfully rolled out
uadmin@ubuntu1:~/devops3/flask_mysql_app$
```

8. Updated website